Estd.1917

Department of CSE, U.C.E, O.U
CLASS TEST – II (MAY, 2026)
B.E (CSE and AI&ML) VI Semester
(Open elective offered by Mechanical Engg. Dept.)
Subject: 3D PRINTING TECHNOLOGY

Estd.1929

Test Duration: 1.5 hours **Maximum Marks: 30**

Note: i. First Question is compulsory and answer any two questions from the remaining.
ii. Missing data, if any, may suitably be assumed

Q. No		Question	Marks	CO No.	BT Level
1	a	**Distinguish** between SLM and EBM 3D Printing Process	2	3	2
	b	**Identify** various types of Non-Manifold Conditions in STL File	2	4	2
	c	**Mention** the features of Mimics software used in 3D Printing	2	4	1
	d	**List** the various types of applications of 3D Printing in automobile sector	2	5	1
	e	**Identify** the applications of 3D Printing technology in electronic industry	2	5	2
2	a	**Illustrate** with a neat sketch the working of SLM and mention its applications	7	3	2
	b	**Compare** between DMLS and LENS	3	3	2
3	a	Briefly **elaborate** on the various STL file problems with a neat sketch	7	4	2
	b	**Recognise** the most appropriate 3D Printing software that can be used for the following (i) Biomodelling of implants (ii) Designing jewellery and (iii) Slicing the STL file	3	4	2
4	a	**Justify** the role of AI & ML in 3D Printing for smarter manufacturing with an example	5	5	3
	b	With an example discuss the **application** of 3D Printing in planning complex surgeries	5	5	3

Note: CO: Course Outcome, BT: Bloom's Taxonomy